

Vector-Based Goal Representation for a Mobile Robot

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Abstract

This project introduces fundamental vector operations used in mobile robot navigation. The positions of a robot and a target are represented as two-dimensional vectors. From these positions, the displacement vector, Euclidean distance, and normalized direction toward the target are calculated. These operations provide the mathematical basis for later motion control, path planning, and obstacle avoidance algorithms.

1 Introduction

The positions of a robot and a target are represented as two-dimensional vectors. From these positions, the displacement vector, Euclidean distance, and normalized direction toward the target are calculated.

The central idea is that a robot needs to know:

- where it is,
- where the goal is,
- how far away the goal is
- and in which direction it should move.

In a two-dimensional environment positions are represented using vectors:

$$\mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix}$$

2 Theoretical Background

2.1 Position Vectors

Define the robot position and goal position as:

$$\mathbf{p}_R = \begin{bmatrix} x_R \\ y_R \end{bmatrix} \quad \mathbf{p}_G = \begin{bmatrix} x_G \\ y_G \end{bmatrix}$$

Where:

- p_R is the position of the robot,
- p_G is the position of the goal.

In the experiment:

$$\mathbf{p}_R = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{p}_G = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

2.2 Vector from the Robot to the Goal

The displacement vector is calculated by subtracting the robot position from the goal position:

$$\mathbf{v}_{RG} = \mathbf{p}_G - \mathbf{p}_R$$

2.3 Euclidean Distance

The distance between the robot and the goal is calculated using the Euclidean norm:

$$d = \|\mathbf{v}_{RG}\| \quad (1)$$

$$d = \sqrt{(x_G - x_R)^2 + (y_G - y_R)^2} \quad (2)$$

2.4 Normalized Direction Vector

The direction toward the goal is obtained by normalizing the displacement vector. It contains direction information independently of distance:

$$\hat{\mathbf{v}}_{RG} = \frac{\mathbf{v}_{RG}}{\|\mathbf{v}_{RG}\|} \quad (3)$$

3 Methodology

The procedure was:

- Define the robot position as a two-dimensional NumPy array.
- Define the goal position.
- Subtract both vectors to obtain the robot-to-goal displacement.
- Calculate the Euclidean distance.
- Normalize the displacement vector.
- Display the resulting values.

4 Results

Figure 1 shows the representation of the robot position, goal position, and displacement vector.

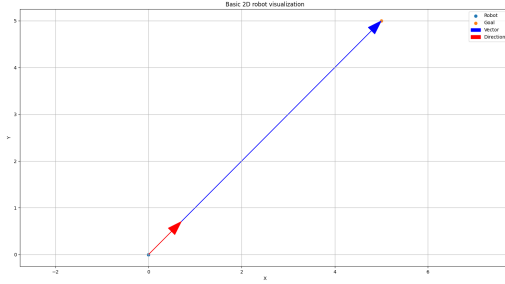


Figure 1: Two-dimensional representation of the robot

$$\text{Distance to goal} = 7.07107, \quad (4)$$

$$\text{Direction to goal} = [0.70711 \quad 0.70711]. \quad (5)$$

5 Project Task

Represent the robot position $\mathbf{p}_R = [0, 0]^T$ and the goal position $\mathbf{p}_G = [5, 5]^T$ as NumPy arrays. Compute the vector from the robot to the goal, determine the Euclidean distance between both positions, and normalize the displacement vector to obtain the direction of motion. Finally, print the calculated values and represent the robot, the goal, and the direction vector in a two-dimensional graph using Matplotlib.